Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/02/2017
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Project Title:	Sleep, Physical Inactivity, Circadian Rhythms and Cardiovascular Vulnerability		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015 NSBRI-RFA-15-01 First Award Fellowships
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Flight Assignment:	NOTE: End date changed to 6/30/2016 per NSBRI (original end date was 9/30/2016) (Ed., 8/18/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shea, Stephen Ph.D. (MENTOR/ Oregon Health & Science University)		
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	POSTDOCTORAL FELLOWSHIP Adverse health outcomes resulting from sleep loss and circadian desynchronization are areas of risk in NASA's Human Research Program (HRP) Roadmap. A stated gap in knowledge (sleep gap #4) is to identify individual vulnerabilities and resiliencies to sleep loss and circadian rhythm disruption to aid individualized countermeasure regimens. Moreover, cardiovascular (CV) gap #8 hypothesizes that the cumulative effect of spaceflight stressors might increase long-term CV disease risk due to factors including but not limited to microgravity, oxidative stress, and alterations in exercise and diet. Additionally, National Space Biomedical Resarch Institute (NSBRI) research themes seek solutions to health concerns facing astronauts that would also benefit healthcare on Earth. Our goal in this ground based study is to identify the endogenous circadian rhythm in endothelial function, a key regulator of CV health, and to study the effects on		

Task Description:	endothelial function of circadian misalignment – as occurs during spaceflight. This study targets risk areas for NASA and fills mechanistic gaps in the HRP, while at the same time contributing the field of CV disease at large. We have one main aim and one exploratory aim. Main Aim: To determine if an endogenous circadian rhythm exists in baseline endothelial function, with worst function at the circadian phase that corresponds to the peak in vulnerability to serious adverse cardiovascular events. Exploratory Aim: To determine if the effects of standardized activities on endothelial function when performed when misaligned (akin to night shift work) result in worse endothelial function compared to when these identical activities are performed across a normally aligned day. (2) Key findings. Our preliminary data suggest that the endogenous circadian system does indeed modulate vascular endothelial function such that when measured as flow mediated dilation, endothelial function is impaired during the biological night with the greatest impairment occurring in the vulnerable morning hours. These data are preliminary and need confirmation in a larger sample size and in different population groups for verification. If these preliminary results hold in a larger sample size and in people with increased cardiovascular risk, we will conclude that the endogenous circadian system is at least in part responsible for impaired vascular function in the vulnerable morning hours and may be involved in the increased risk of adverse cardiovascular events in this period. (3) Impact of key findings on hypotheses, technology requirements, objectives and specific aims of the original proposal. Our preliminary key findings are in support of our main aim. We are still continuing to study more volunteers and assay the biomarkers. Blood biomarkers will inform us about the mechanisms that may drive the endogenous circadian rhythm in vascular endothelial function. (4) Proposed research plan for the coming year. We plan on publishing our full results in the upcoming year, and currently are running a protocol aimed in separating out the effects of physical inactivity and sleep on impaired vascular function in the morning. ORIGINAL INFORMATION OCTOBER 2015 WHEN NEW PROJECT ESTABLISHED: Round-the-clock work schedules aboard spaceships and in most industries on Earth render people to increased risk for developing cardiovascular diseases (CVD), in part via the disruption of endogenous human clocks. Endogenous circadian rhythms, generated by these clocks form the basic framework of human physiology. In order to improve our understanding of the mechanisms leading to increased CVD, it is essential to understand the endogenous circadian rhythm in vascular endothelial function, which is an early and a robust marker of CVD. We will determine if an endogenous circadian rhythm in endothelial function exists using a 5 day forced desynchrony protocol. We will also determine if standardized activities performed during the night shift impair endothelial function. Twelve healthy participants will live on four artificial 18 hour days consisting of six hour sleep and 12 hour wake periods. Participants will be isolated from all time cues and will remain in dim light during the wake periods and complete darkness during sleep. The dim light allows the human pacemaker to freely run. All behaviors including food and activity will be identical across all wake periods such that the effects of the behaviors and the endogenous circadian clocks can be separated out. Endothelial function will be measured as flow mediated dilation (FMD) every four hours during the wake periods. To understand mechanisms, blood will be drawn to measure physiological markers such as epinephrine, norepinephrine, cortisol, oxidative stress and inflammatory cytokines. Biomarkers inform us about the mechanisms modulating vascular function at various circadian phases. Identification of a rhythm in endothelial function along with the mechanisms that control it will set the stage for developing chronotherapeutic countermeasures against vascular function impairment (such as timing antioxidants and exercise) in space and on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Adverse cardiovascular events on Earth occur most commonly during the morning hours immediately upon awakening. We are finding that the impaired vascular endothelial function in the morning can at least in part be explained by the circadian profile of vascular endothelial function. If these preliminary results hold in a larger sample size and in people with increased cardiovascular risk, we will conclude that the endogenous circadian system is at least in part responsible for impaired vascular function in the vulnerable morning hours and may be involved in the increased risk of adverse cardiovascular events in this period. Further, we are currently investigating the effects of behaviors such as prior sleep and prolonged inactivity on impaired vascular function in the morning. Understanding the interactive effects of these variables on morning endothelial function will allow us to design experiments to test countermeasures against impaired endothelial function thereby reducing cardiovascular risk.
Task Progress:	Round-the-clock work schedules aboard spaceships and in most industries on Earth render people to increased risk for developing cardiovascular diseases (CVD), in part via the disruption of endogenous human clocks. Endogenous circadian rhythms, generated by these clocks, form the basic framework of human physiology. In order to improve our understanding of the mechanisms leading to increased CVD, it is essential to understand if there is an endogenous circadian rhythm in vascular endothelial function, which is an early and a robust marker of CVD. We are investigating if an endogenous circadian rhythm in endothelial function exists using a 5 day forced desynchrony protocol. We are also determining if standardized activities performed during the night shift impair endothelial function. In order to separate out the effects of the behavior from that of the circadian system, we have studied 10 healthy participants who lived on four artificial 5 h 20 m days consisting of sleep and wake periods of 2 h 40 m each during which all activities were standardized. (We note that 8 healthy participants were completed by the time the NSBRI fellowship was relinquished due to receipt of separate NIH funding.) Additionally, participants were isolated from all time cues and remained in dim light during the wake periods and complete darkness during sleep. All behaviors including food, and activity were scheduled with exact timing across all wake periods such that the effects of the behaviors and the endogenous circadian clocks can be separated. Endothelial function was measured as flow mediated dilation (FMD) under standard conditions during each wake period. To understand mechanisms, blood was drawn to measure physiological markers, including epinephrine, norepinephrine, cortisol, oxidative stress, and inflammatory cytokines. Dim light melatonin was measured to calculate the circadian period for each participant, and phase for each measurement. Our preliminary results suggest that vascular endothelial function is at least in part modulated by the endogenous circadian system such that vascular function is low during the biological night and extending across the vulnerable

	morning hours. These data are preliminary and need confirmation in a larger sample size and in different population groups for verification. If these preliminary results hold in a larger sample size and in people with increased cardiovascular risk, we will conclude that the endogenous circadian system is at least in part responsible for impaired vascular function in the vulnerable morning hours and may be involved in the increased risk of adverse cardiovascular events in this period. Identification of a rhythm in endothelial function along with the mechanisms that control it will set the stage for developing chronotherapeutic countermeasures against vascular function impairment (such as timing antioxidants and exercise) in space and on Earth.
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